

# VEDANT BAJAJ

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## PROFESSIONAL EXPERIENCE

### Data Engineer, Co-Op | Conifer Health Solutions, USA

Aug. 2025 – May. 2026

#### SFTP-to-GCS Healthcare Data Pipeline

- Engineered a reliable, scalable, and observable Airflow pipeline to ingest 27 recurring 400 MB–2 GB healthcare files from SFTP to GCS across 450+ facilities, using multi-level audit tables for validation, lineage, status tracking, and error recovery.
- Built a serverless Cloud Run loading service and modular Bronze, Silver, and Gold architecture in BigQuery using dbt and parameterized stored procedures, enabling reusable transformations and fault-isolated processing
- Implemented automated source-to-target reconciliation for row counts, aggregates, nulls, duplicates, schema consistency, and business rules, reducing reporting discrepancies by 30% and strengthening trust in downstream data.
- Optimized partitioned and clustered Gold-layer models and analytical SQL to deliver consistent business metrics with 45% faster query performance, while audit-driven troubleshooting reduced incident-resolution time by 40%.

#### DB2-to-Bigquery Migration

- Modernized a large-scale legacy data platform by migrating 800+ DB2 tables and 650+ stored procedures to BigQuery, using dependency analysis, schema mapping, and cloud-native redesign to improve scalability and maintainability.
- Reengineered procedural DB2 logic into reusable BigQuery Standard SQL with partitioning, clustering, and optimized set-based transformations, improving query performance by 45% and reducing unnecessary data processing.
- Used Vertex AI to accelerate repetitive SQL conversion and test generation, while enforcing human review, access controls, and production validation to deliver accurate, governed, and analytics-ready datasets.

### Software Engineer | Rabbit and Tortoise Technology Solutions, India

Nov. 2022 – Sep. 2023

- Engineered a normalized relational data model spanning 6 core business domains—products, invoices, customers, suppliers, payments, and inventory—enabling consistent reporting and real-time stock tracking across 1,000+ SKUs.
- Developed automated Excel-to-database ingestion pipelines for bulk product loads, incorporating schema validation, duplicate detection, and ACID-compliant updates to prevent partial writes and preserve inventory accuracy.

### Software Engineer, Intern | Bizsol IT Services, India

Aug. 2021 – Dec. 2021

- Engineered SQL-based reconciliation pipelines using complex joins, aggregations, and rule-driven validations to identify discrepancies across taxation datasets, strengthening data quality and audit readiness.
- Optimized recurring reconciliation workloads through query refactoring, indexing, and efficient data-access patterns, improving execution performance and reliability for financial reporting cycles.

## PROJECTS

### Healthcare Claims Data Platform | <https://github.com/VedantBajaj/healthcare-claims-data-platform>

- Built an end-to-end healthcare claims platform for 900K+ CMS SynPUF records using Python, PostgreSQL, Docker, and Bronze, Silver, and Gold layers, creating a reliable foundation for claims analytics.
- Developed modular dbt models and 12 automated quality checks for duplicates, nulls, referential integrity, negative payments, and source-to-target reconciliation, stopping bad data before it reached reporting.
- Orchestrated daily pipelines with Apache Airflow, monitored freshness, failures, and claim KPIs in Grafana, and used AI-assisted SQL and test generation to speed development and improve model accuracy.

### Real-Time Financial Sentiment Analysis Platform | <https://github.com/VedantBajaj/sentiment-platform>

- Engineered a near real-time financial news platform using Kafka, Spark Structured Streaming, Python, and TimescaleDB to continuously process live events and enrich every record with 2 AI-driven outputs: sentiment scores and named entities.
- Built a scalable, fault-tolerant streaming architecture with Kafka partitions, consumer groups, Spark checkpoints, and idempotent writes, enabling reliable recovery and consistent analytics under continuous event flow.
- Containerized 100% of the 4-service platform—Kafka, Spark, TimescaleDB, and Grafana—with Docker Compose and monitored 4 operational dimensions: throughput, freshness, failures, and sentiment trends.

## TECHNICAL SKILLS

**Languages:** Python, SQL, JavaScript

**Data Engineering:** ETL/ELT Pipelines, Batch and Streaming Processing, Data Warehousing, Dimensional Data Modeling, Medallion Architecture, Incremental Loads, Source-to-Target Reconciliation, Data Quality, Schema Validation, Pipeline Reliability

**Cloud & Orchestration:** Google Cloud Platform, BigQuery, Google Cloud Storage, Cloud Run, Google Workflows, Apache Airflow

**Big Data & Streaming:** Apache Kafka, Spark Structured Streaming, Apache Spark, Event-Driven Architecture, Checkpointing, Fault-Tolerant Processing

**Databases & Analytics:** PostgreSQL, SQL Server, DB2, MySQL, TimescaleDB, Query Optimization, Partitioning, Clustering

**Transformation & Observability:** dbt, Docker, Grafana, Pandas, NumPy, Git, Workflow Monitoring, Data Freshness, Pipeline Logging, Performance Tuning

**AI-Assisted Engineering:** Vertex AI, AI-Assisted SQL Development, Validation Test Generation, Edge-Case Analysis, NLP

## EDUCATION

**University of Texas at Dallas | Master of Science in Computer Engineering**

Aug. 2024 – May. 2026

**University of Pune | Bachelor of Technology in Computer Engineering**

Aug. 2018 – May. 2022